PET361 Milestone 1

15 September 2025

Keatlegile Rachidi
603127 Belgium Campus

Table of Contents

1.1 The virus can delete the files in the root directory.......... 2

1.2 The steps taken before the attack and after the attack is clear and completely documented........ 3

Before Attack 3

In the time of the attack 3

After the attack 4

1.3 Proof that Kali Linux VM does not work 5

1.4 Program runs without having to open a code editor.......... 6

References 7

PET361 Milestone 1

This is a bash script that destroys a Kali Linux VM by recursively deleting the root file system, it can be executed by double-clicking on the file "nuke-system.sh" as a root user with the right permissions, and the file permissions need to be set to executable.

1.1 The virus can delete the files in the root directory.

Command: sudo rm -rf –no-preserve-root /

1. The Shebang Line (#!/bin/bash)
    - This is a "hashbang" line
    - #! tells the Linux kernel this is an executable script
    - /bin/bash specifies which interpreter to use (the Bash shell)
    - When you run the script, Linux reads this line first and knows to execute the file using /bin/bash

Sudo: Executes the command with root privileges.

rm: The remove command.

-r: Recursively deletes directories and their contents.

-f: Forces the deletion, ignoring non-existent files and never prompting for confirmation.

--no-preserve-root: This crucial option bypasses the safety mechanism that prevents accidental deletion of the root directory.

/: Specifies the root directory as the target.

1.2 The steps taken before the attack and after the attack is clear and completely documented.

Before Attack

As seen above:

- The malicious script requires root access.
- The root directory contains all the necessary system files and directories.
- The Kali VM is fully functional at this point.

In the time of the attack

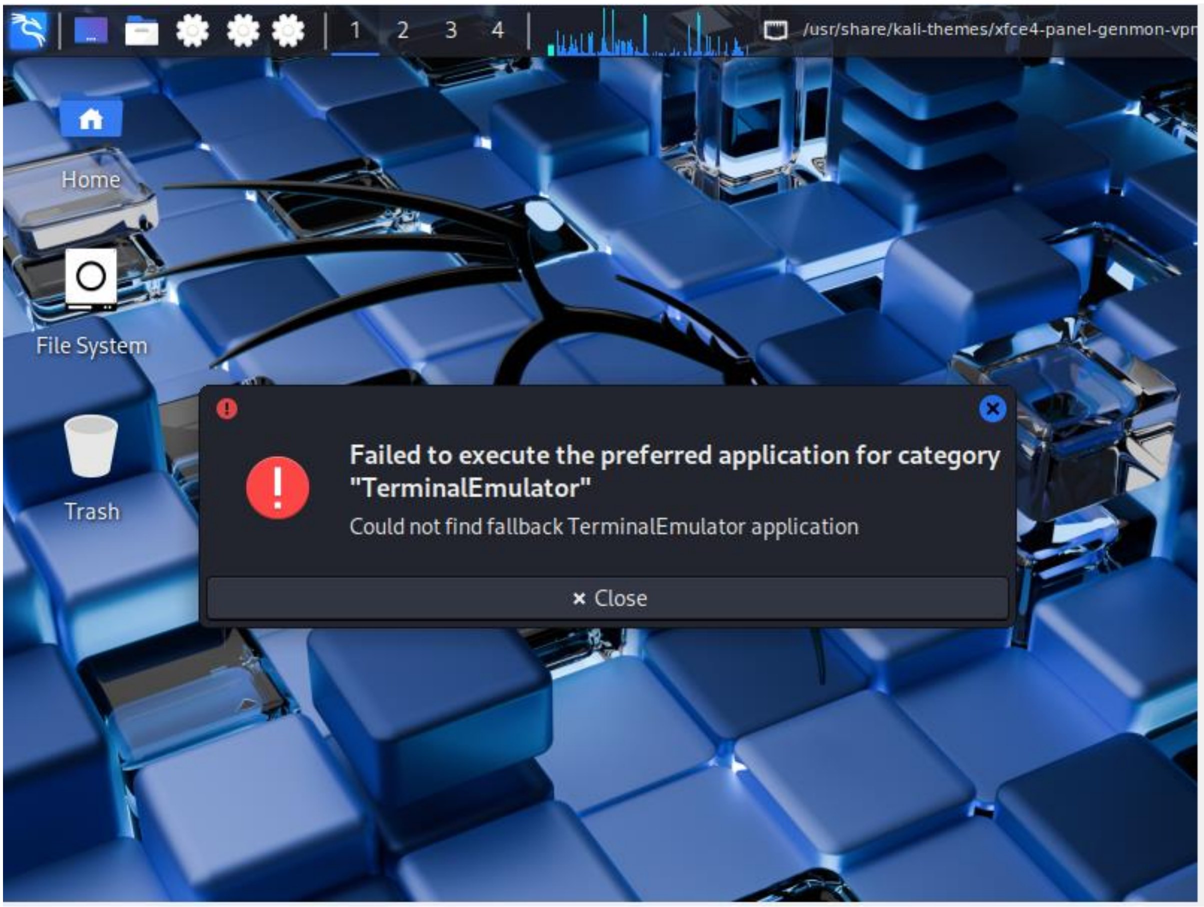
If the code was active:

- “Sudo” will run the command with root privileges and has full permission to delete everything.
- “rm” will delete files and directories
- “-r” will recursively delete directories and their contents.
- “-f” will force deletion, ignoring confirmation.
- “—no-preserve-root”: by default “rm -rf /” is blocked to protect the user from deleting the entire system. This command disables that safeguard, allowing deletion of the root directory itself and everything inside of it.
- You will immediately notice that the program starts running with the current user’s privileges and in this case, it is ran by sudo or the admin, anything can be deleted.
- Programs will start to freeze and the terminal will stop responding. At this point, files are being deleted and running processes may start to crash.

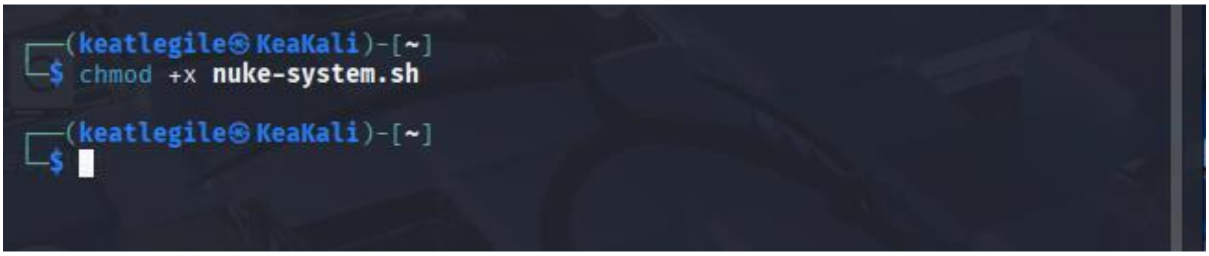
After the attack

- If the deletion completes, the machine will be unbootable since important data and OS files will be missing. User data may be permanently lost.
- The system is unstable. Some applications might run until a reboot while other will not.
- The system is partially stable but broken and commands such as ls, sudo or even bash may no longer work.
- The data in /home and hidden directories will be gone.

1.3 Proof that Kali Linux VM does not work



1.4 Program runs without having to open a code editor



To run a program without opening a code editor, an executable script file is created. This is done using the “chmod +x” command, which grants the necessary execute permissions to the file.

chmod +x nuke-system.sh

chmod = this command is used to modify permissions.

+x = this adds execute permissions for all users.

References

Gopinathan, A. (2021). Deleting the Root Directory in Your Linux OS. [online] Geek Culture. Available at: https://medium.com/geekculture/deleting-the-root-directory-in-your-linux-os-8f38e3add4f6.

GeeksforGeeks (2021). rm rf Command in Linux With Examples. [online] GeeksforGeeks. Available at: https://www.geeksforgeeks.org/linux-unix/rm-rf-command-in-linux-with-examples/ [Accessed 13 Sep. 2025].

Ibm.com. (2023). AIX. [online] Available at: https://www.ibm.com/docs/en/aix/7.1.0?topic=files-deleting-rm-command.

sudo (2016). Unix & Linux Stack Exchange. [online] Unix & Linux Stack Exchange. Available at: https://unix.stackexchange.com/questions/302468/sudo-rm-rf-no-preserve-root-vs-sudo-rm-rf [Accessed 13 Sep. 2025].

in (2018). Unix & Linux Stack Exchange. [online] Unix & Linux Stack Exchange. Available at: https://unix.stackexchange.com/questions/437111/what-would-happen-if-i-did-rm-rf-in-the-root-directory [Accessed 14 Sep. 2025].

The Linux Cast (2022). What Happens if You Really Delete Your Root Directory? [online] YouTube. Available at: https://www.youtube.com/watch?v=dAITrQ8usl8 [Accessed 14 Sep. 2025].

Robbins, A. (2006a). Unix in a Nutshell. O'Reilly Media, Inc.